How to enter values into the RMTI equations

Step-by-step guide for the time-indexed RMTI with oncoclastics, with worked examples

Read this first. Every number in this guide is a **sample**. Example 1 uses inputs already in the repository (the six published HCC cohorts of the HCC-surveillance preprint, *Alcohol cirrhosis*); Examples 2 and 3 use made-up values for an imaginary patient so that each step can be shown. Nothing here is validated against outcomes, no tier is a diagnosis, prognosis or treatment recommendation, and the RMTI is a research framework, not a medical device. Replace the sample values with your own only after deciding where each one comes from (Part 1).

What you will do. Score five inputs (L, E, V, ρ, C), multiply them in the order the equations require, read the tier, and then add the two senses of oncoclastics (a biological and a care-system part of ρ) and the time-indexed version. Each step shows the formula, the sample numbers in it, and the result. All figures were produced by the repository's own code and checked against it.

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1 The equations on one page

Symbol	Meaning	Range	Direction	Where the value comes from
L	Likelihood of the outcome over a stated horizon	0 – 1	Raises R	A stage-anchored or validated model, rescaled to 0–1 (HCC surveillance application: published incidence mapped to 0.15–0.90)
E	Exposure: burden borne (individual) or share of the population in the footprint (programme)	0 – 1	Raises R	Rubric levels (manuscript Table S4: 0.10, 0.30, 0.50, 0.70, 0.90) or, for a programme, the share inside the surveillance footprint (0.55–0.70)
V	Vulnerability: accumulated cost of prior injury	0 – 1	Raises R	HCC application: ALBI grade 1 / 2 / 3 → 0.35 / 0.60 / 0.85. Time-indexed: from an allostatic-load indicator
ρ	Resilience: the fraction of inherent risk that is absorbed	0 – 0.99	Lowers R, through (1 − ρ)	Care-system part ρ_{care}: surveillance ladder. Biological part ρ_{bio}: five anchor levels. Combine (Eq. 4)
C	Consequence if the outcome occurs (used only in EAL)	0.10 – 1.00	Raises EAL	Integer 1–5 → 0.10, 0.30, 0.50, 0.75, 1.00; HCC: BCLC 0, A, early-intermediate, B, C–D

The equations (numbered for this guide; the repository's code comments use their own numbers)

```
(1)  IR  = L × E × V                          inherent risk
(2)  R   = IR × (1 − ρ)                       residual risk; tier read from R
(3)  EAL = L × C × (1 − ρ)                    expected annual loss, C in 0.10–1.00
(4)  ρ   = 1 − (1 − ρ_bio) × (1 − ρ_care)     two senses combined; ρ_bio = 0 gives ρ = ρ_care
(5)  % cut in R = 1 − (1 − ρ_after) / (1 − ρ_now)       holding IR fixed
(6)  IEI = (% cut as a fraction) / (1 − ρ_after)         above 1: more risk removed than left behind
```

Direction matters. A *higher* ρ always *lowers* R. L, E and V multiply, so a low value in any one of them lowers inherent risk sharply. ρ must be below 1 (the Python core rejects 1 or more; all inputs in the tools stop at 0.99).

Tiers (fixed thresholds on R, read from the unrounded product):

Tier	R is	Tier	R is
1 · Low	below 0.05	4 · Severe	0.35 up to 0.60
2 · Moderate	0.05 up to 0.15	5 · Critical	0.60 or more
3 · High	0.15 up to 0.35		

Rounding. R, IR and EAL are displayed to 3 decimals, rounded half-up from the exact decimal product (not the binary float), so 0.30 × 0.65 × 0.60 × 0.50 shows 0.059. The tier is read from the unrounded R.

The three ladders for ρ (advisory reference points, not a lookup; a value between two points is allowed):

Ladder	Used for	Reference points
HCC surveillance effectiveness	ρ_{care} (care-system)	none 0.15, usual 0.20, outreach 0.45, specialist 0.50, optimised 0.65
Biological adaptive reserve (provisional, assigned, not validated)	ρ_{bio} (biological)	not scored 0.00, little 0.10, limited 0.25, moderate 0.40, good 0.55, excellent 0.70
Disaster-risk coping capacity	Disaster applications	none 0.10, ad hoc 0.20, monitoring enforcement 0.25, structured 0.45, structured to funded 0.55, funded 0.58, optimised 0.70

2 Example 1: one published cohort, step by step

Alcohol cirrhosis is one of the six published cohorts in the repository (HCC-surveillance preprint, Table 2; also in the HCC workbook). Its inputs are L = 0.30, E = 0.65, V = 0.60, ρ now = 0.20, ρ after a surveillance improvement = 0.50, C = 5. They are shown here to teach the arithmetic, not as a recommendation.

Step 1. Enter L, E and V

Each is a number from 0 to 1. Check the direction: bigger means more inherent risk.

```
L = 0.30     E = 0.65     V = 0.60
```

Step 2. Multiply them: inherent risk (Eq. 1)

Multiply the three numbers. No resilience yet.

```
IR = L × E × V = 0.30 × 0.65 × 0.60 = 0.117
```

Step 3. Enter ρ (now)

Pick the resilience that matches how surveillance is delivered today. Here the ladder point *usual* care is 0.20. Resilience is the fraction of IR that is absorbed, so 0.20 means 20% is absorbed and 80% is left.

```
ρ now = 0.20        (1 − ρ) = 1 − 0.20 = 0.8
```

Step 4. Residual risk R (Eq. 2)

Multiply IR by the unabsorbed fraction.

```
R = IR × (1 − ρ) = 0.117 × 0.8 = 0.0936   →  shown as 0.094
```

Step 5. Read the tier

Compare the unrounded R with the thresholds in Part 1. R = 0.0936 falls in **Tier 2 - Moderate**.

Step 6. Enter C and compute EAL (Eq. 3)

C is an integer from 1 to 5 that maps to a value: 1 → 0.10, 2 → 0.30, 3 → 0.50, 4 → 0.75, 5 → 1.00. For HCC it is the expected stage at detection (BCLC 0, A, early-intermediate, B, C–D). Here C = 5, value 1.00.

```
EAL = L × C × (1 − ρ) = 0.30 × 1.00 × 0.8 = 0.24   →  shown as 0.240
```

EAL is a relative index for ranking, not a count of patients or a probability of loss.

Step 7. What if surveillance improved? (ρ after)

Keep L, E, V and C. Change only ρ to the improved value, 0.50.

```
R after = 0.117 × (1 − 0.50) = 0.117 × 0.5 = 0.0585   →  Tier 2 - Moderate
% cut in R (Eq. 5) = 1 − (1 − 0.50) / (1 − 0.20) = 1 − 0.5 / 0.8 = 37.5%
IEI (Eq. 6) = 0.375 / (1 − 0.50) = 0.750
```

The six published cohorts through the same steps (computed by the repository code; same inputs as the HCC workbook):

Cohort	L	E	V	ρ now	ρ after	C	R now	Tier now	R after	Tier after
Untreated HBV	0.90	0.70	0.55	0.10	0.45	5	0.312	High	0.191	High
Active viral cirrhosis	0.55	0.70	0.53	0.50	0.65	5	0.102	Moderate	0.071	Moderate
Alcohol cirrhosis	0.30	0.65	0.60	0.20	0.50	5	0.094	Moderate	0.059	Moderate
Cured-HCV cirrhosis	0.25	0.60	0.48	0.20	0.55	5	0.058	Moderate	0.032	Low
Suppressed-HBV	0.20	0.55	0.45	0.20	0.55	5	0.040	Low	0.022	Low
NAFLD cirrhosis	0.15	0.60	0.53	0.20	0.50	5	0.038	Low	0.024	Low

3 Example 2: one patient, both senses of ρ (oncoclastics)

Oncoclastics applies hazard–exposure–vulnerability–resilience assessment to the fragmentation of cancer-related trajectories, in two senses that are always named: **biological** (progression from chronic injury toward cancer) and **care-system** (disruption of surveillance and care). In the equations the care-system sense enters *only* through ρ. The patient below is imaginary, and every value is a sample.

Step 1. Enter L and E

L comes from a stage-anchored or validated model, rescaled to 0–1; E from the rubric (Table S4: 0.50 = moderate burden, partly controlled).

```
L = 0.30     E = 0.50
```

Step 2. Get V from the ALBI grade

In the HCC application V comes from the ALBI grade. ALBI is computed from serum albumin (g/L) and total bilirubin (µmol/L); grade 1 is at or below −2.60, grade 2 is above −2.60 up to −1.39, grade 3 is above −1.39 (Johnson et al., 2015). Check your laboratory's units before using the formula.

```
ALBI = 0.66 × log10(bilirubin) − 0.085 × albumin
     = 0.66 × log10(17) − 0.085 × 38 = -2.418   →  grade 2
V = 0.60        (grade 1 → 0.35, grade 2 → 0.60, grade 3 → 0.85)
```

Step 3. Enter ρ_{care} (care-system)

Pick the point on the surveillance-effectiveness ladder that matches how surveillance is actually delivered: none 0.15, usual 0.20, outreach 0.45, specialist 0.50, optimised 0.65. Values in between are allowed. Surveillance delivery, access, follow-up and adherence belong here and **nowhere else**.

```
ρ_care = 0.20   (usual care)
```

Step 4. Enter ρ_{bio} (biological) from the anchor ladder

Score biology only, from evidence that is **not already used for another term**. Here ALBI already set V, so ρ_bio is placed on other evidence (sample: impaired metabolic flexibility, partial recovery after exposure was reduced), which is the “moderate” anchor. If you have no such evidence, leave ρ_bio at 0.00 (not scored): the result is then the standard score.

```
Anchors: not scored 0.00, little 0.10, limited 0.25, moderate 0.40, good 0.55, excellent 0.70
ρ_bio = 0.40   (moderate; the sandbox slider stops at 0.70)
```

The five ρ_bio anchors are provisional and assigned, not estimated or validated. Do not read 0.70 as a measured 70% absorption.

Step 5. Combine the two senses (Eq. 4)

Both parts absorb risk, but the same risk cannot be absorbed twice. The formula multiplies the unabsorbed fractions, which keeps ρ below 1.

```
ρ = 1 − (1 − ρ_bio) × (1 − ρ_care) = 1 − (1 − 0.40) × (1 − 0.20) = 1 − 0.6 × 0.8 = 0.520
(simple addition would give 0.60, which double-counts)
```

Step 6. Compute IR, R, tier and EAL (Eqs. 1–3)

Exactly as in Example 1, but with the combined ρ.

```
IR  = 0.30 × 0.50 × 0.60 = 0.09
R   = 0.09 × (1 − 0.520) = 0.09 × 0.480 = 0.0432   →  0.043, Tier 1 - Low
EAL = 0.30 × 0.30 × 0.480 = 0.0432   (C = 2, BCLC A, value 0.30)
```

C stays at its biological value: the effect of care enters through ρ_care only, and C is not re-scored when care changes. If you want to see the alternative, run it as a separate, pre-specified sensitivity analysis.

Step 7. What does each part add?

Remove one part at a time and keep the other. These marginal effects are conditional and do not add up to the total absorbed risk.

```
Without care  (ρ = ρ_bio = 0.40):  R = 0.054   →  care lowers R by 0.0108
Without biology (ρ = ρ_care = 0.20): R = 0.072   →  biology lowers R by 0.0288
```

Step 8. Model a disruption of care

Keep the biology (L, E, V, ρ_bio) and lower only ρ_care, for example to 0.05 because follow-up lapses. The probability of a disruption is not modeled; a disruption is a fall in ρ_care.

```
ρ = 1 − (1 − 0.40) × (1 − 0.05) = 0.430
R = 0.09 × (1 − 0.430) = 0.0513   →  change in R = 0.0081   (Tier 1 - Low → Tier 2 - Moderate)
```

4 Example 3: the time-indexed version (V and ρ updated each cycle)

Instead of typing V and ρ once, you supply, for every six-monthly surveillance cycle t, the allostatic-load indicator y(t) (0 to 6; a missed visit is left blank) and the adherence, also called delivery in the code (the share of planned surveillance delivered, 0 to 1). The filter turns these into V*(t) and ρ(t), and the core then scores each cycle. All gains are **uncalibrated placeholders**: α = 0.41, β = 0.09, look-ahead κH = 1, λ = 1, V from 0.35 (y = 0) to 0.85 (y = 6).

Sample inputs, same imaginary patient as Example 2 (L = 0.30, E = 0.50, C = 2, ρ_bio = 0.40, ρ_care at the start = 0.20):

Cycle t	0	1	2	3
y(t), allostatic load	2	3	3	4
adherence(t) = delivery(t)	0.40	0.40	0.30	0.30

Step 1. Start the filter at cycle 0

Level starts at the first observation and slope at 0.

```
level(0) = y(0) = 2     slope(0) = 0
```

Step 2. Update at cycle 1: predict, then correct

Predict with the previous level plus slope; the innovation is the surprise; then move the level by α times it and the slope by β times it. (A missed visit would carry the prediction forward.)

```
level_pred = level(0) + slope(0) = 2 + 0 = 2
innovation  = y(1) − level_pred = 3 − 2 = 1
level(1) = 2 + 0.41 × 1 = 2.41
slope(1) = 0 + 0.09 × 1 = 0.09
```

Step 3. Turn the state into V*(t)

Add the look-ahead (κH × slope) to the level, divide by the top of the scale, and map onto the V range.

```
drive = level + κH × slope = 2.41 + 1 × 0.09 = 2.50
V*(1) = 0.35 + (0.85 − 0.35) × 2.50 / 6 = 0.5583
```

Step 4. Update ρ from delivery

ρ_care moves by λ times the change in delivery since the previous cycle, and is kept between 0 and the ceiling 0.95. Then combine with ρ_bio as before.

```
ρ_care(1) = ρ_care(0) + λ × (delivery(1) − delivery(0)) = 0.20 + 1 × (0.40 − 0.40) = 0.20
ρ(1) = 1 − (1 − 0.40) × (1 − 0.20) = 0.520
```

Step 5. Score the cycle through the core

The same Eqs. 1–2 with V*(t) and ρ(t).

```
R(1) = L × E × V*(1) × (1 − ρ(1)) = 0.30 × 0.50 × 0.5583 × 0.480 = 0.04020
```

All four cycles (repeat steps 2–5 for each cycle; computed by the repository code):

t	y	delivery	level	slope	V*	ρ_care	ρ (both)	R	Tier
0	2	0.40	2.000	0.000	0.5167	0.200	0.520	0.0372	Low
1	3	0.40	2.410	0.090	0.5583	0.200	0.520	0.0402	Low
2	3	0.30	2.705	0.135	0.5867	0.100	0.460	0.0475	Low
3	4	0.30	3.316	0.239	0.6463	0.100	0.460	0.0523	Moderate

Candidate oncoclastic events in this run (provisional, descriptive flags, not alerts): t = 1: V* rose (biological); t = 2: delivery fell (care-system); t = 2: V* rose (biological); t = 3: V* rose (biological). A missed visit or a fall in delivery is a care-system event; a rise in V* is a biological event. The event rules are uncalibrated.

> With ρ_bio = 0 this table is exactly the existing HCC time-indexed result, because the existing ρ(t) is the care-system part. Whether a trajectory adds information beyond a well-measured current level is an open question, not a finding.

5 Where to type the values

A. Python (no installation beyond Python 3.8; run from the repository root):

```
import sys; sys.path.insert(0, "rmti-python"); sys.path.insert(0, "oncoclastics")
from core.rmti_core import score
from rmti_oncoclastics import score_two_senses, combine_rho, RHO_BIO_ANCHORS
score(0.30, 0.65, 0.60, 0.20, 5)                          # L, E, V, rho, C  (Example 1: Eqs. 1-3)
rho = combine_rho(rho_bio=0.40, rho_care=0.20)             # Eq. 4 -> 0.52
score_two_senses(0.30, 0.50, 0.60, 0.40, 0.20, 2)          # L, E, V, rho_bio, rho_care, C
                                                           # both senses + marginal effects
```

B. Oncology sandbox (`oncology/index.html`, or the GitHub Pages page). Use the oncology tab.

Field on the page	Type or choose	Note
Likelihood (L), Exposure (E), Vulnerability (V)	Sliders, 0–1	Step 0.01
Resilience now (ρ)	Slider, 0–0.99	In the oncoclastics panel this is read as ρ_care
Resilience after surveillance improvement (ρ)	Slider, 0–0.99	Gives R after and the % cut
Consequence if realized (C)	Choose the stage	Maps to 1–5 and then 0.10–1.00
Oncoclastics panel: Biological adaptive reserve (ρ_bio)	Click an anchor or slide, 0–0.70	0.00 = not scored = standard score
Oncoclastics panel: If care capacity moved to (ρ_care)	Slider, 0–0.99	Starts at half of ρ now; a value below ρ now is a disruption of care
Trajectory panel: Allostatic load (0–6) and Adherence (0–1)	One row per six-monthly cycle, t = 0, 1, 2…	Fill rows in order with no gaps; leave allostatic load blank for a missed visit

C. HCC workbook (`calculators/RMTI_HCC_Calculator_v2.3.0.xlsx`, sheet *RMTI Calculator*). Type only in the shaded input cells, one row per cohort or patient (rows 5–56):

Column	Enter	Allowed
A, B, C	Cohort or patient name, source, aetiology note	text

Column	Enter	Allowed
D	N, number at risk	a number above 0
E, F, G	L, E, V	decimal between 0 and 1
H, I	ρ now, ρ after	decimal between 0 and 0.99
J	C as an integer	1 to 5
K to Y	Calculated: C value, L·E·V, R now, tier now, R after, tier after, change in risk, EAL, priority, IEI, posture text	do not overwrite

The workbook has no ρ_bio column. To use both senses there, compute ρ with Eq. 4 first and type the result in column H or I. The *Trajectory* sheet takes the cycle-by-cycle indicator and adherence values in its input cells (blue text); its model parameters are in cells B5 to B13.

6 Checks and common mistakes

Mistake	What to do instead
Entering ρ as if it were risk (high ρ for a bad situation)	ρ is resilience: higher ρ lowers R. A poor surveillance programme has a low ρ (ladder point “none”, 0.15).
Using ρ = 1.00	Not allowed (R would be zero; the core rejects it). Stay at 0.99 or below; ρ_bio stops at 0.70.
Adding ρ_bio and ρ_care	Combine with Eq. 4. 0.40 and 0.20 give 0.52, not 0.60.
Counting the same information twice	Give each measure to one term. Information already used in L must not be reused in E, V or ρ; if ALBI sets V, it must not also place ρ_bio; surveillance delivery is ρ_care only.
Putting surveillance, access or adherence into ρ_bio	Those are ρ_care. ρ_bio is biology only.
Re-scoring C when care changes	Default: care effects enter through ρ_care only. Treat a C change as a separate sensitivity analysis.
Reading an anchor as a measurement	Anchors are ordinal reference points; the ρ_bio anchors are assigned and unvalidated.
Reading a tier as advice	A tier is a relative, unvalidated score. It is not a diagnosis, a prognosis or a treatment recommendation.
Comparing R between populations	R and EAL are relative indices on bounded scales, not probabilities or counts.

Quick self-check. Recompute Example 2 with ρ_bio = 0: the result must equal Example 2 with ρ = ρ_care alone. The repository's tests check this for the Python module, and the sandbox check compares the panel with Python, ρ_bio = 0 included.

Status. Research prototype. Not validated against real outcomes; not a medical device; no clinical action is attached to any score or tier. Sample values were chosen to show arithmetic. The gains of the time-indexed filter, the ρ_bio anchors, the event rules, the independence assumption in Eq. 4 and the treatment of C are all untested. The probability that care is interrupted is not modeled.

7 Blank worksheet (print and fill in)

Write your own values and the source of each. Keep the evidence behind every placement.

Input	Value	Where it comes from (model, measure, ladder point, date)
L likelihood, 0–1		
E exposure, 0–1		
V vulnerability, 0–1 (ALBI grade → 0.35 / 0.60 / 0.85)		
ρ_care care-system, 0–0.99		
ρ_bio biological, 0 to 0.70 (0 = not scored)		
C consequence, integer 1–5		

```
Eq. 4  ρ = 1 − (1 − ρ_bio)(1 − ρ_care) = 1 − (1 − ____)(1 − ____) = ______
Eq. 1  IR = L × E × V = ____ × ____ × ____ = ______
Eq. 2  R = IR × (1 − ρ) = ______ × ______ = ______      Tier:  1   2   3   4   5
       (tier cut-points on R: 0.05 / 0.15 / 0.35 / 0.60)
Eq. 3  EAL = L × C × (1 − ρ) = ____ × ____ × ______ = ______

What if ρ_care moved to ____ :  ρ = ______  R = ______  change in R = ______
Without care (ρ_care = 0): R = ______     Without biology (ρ_bio = 0): R = ______
```

Cycle t	y(t)	adherence	level	slope	V*	ρ_care	ρ (both)	R	Tier
0									
1									
2									
3									
4									
5									

Per cycle: level_pred = level + slope; innovation = y − level_pred; level = level_pred + α × innovation; slope = slope + β × innovation; V* = 0.35 + 0.50 × (level + slope) / 6; ρ_care = ρ_care(previous) + λ × (delivery − previous delivery); then Eqs. 4, 1 and 2.

8 Sources

Equations, ladders and tiers: the repository's `core/rmti_core.py`, `oncology/rmti_hcc.py`, `dynamic/rmti_dynamic.py` and `oncoclastics/rmti_oncoclastics.py`. **Published cohort inputs:** Marolla (2026b), the HCC-surveillance preprint, Table 2. **Rubric levels and the combined ρ:** the accompanying manuscript, Supplementary Material S4 and S5. **Anchors for ρ_bio:** `oncoclastics/RHO_BIO_ANCHORS.md`. **Definition:** `oncoclastics/TERMINOLOGY_NOTE.md`.

Johnson PJ, Berhane S, Kagebayashi C, et al. Assessment of liver function in patients with hepatocellular carcinoma: a new evidence-based approach, the ALBI grade. J Clin Oncol 2015;33:550–558.

Marolla C. (2018). Information and Communication Technology for Sustainable Development. CRC Press. · Marolla C. (2026a). From Narrative to Algorithm: a Risk Mechanism Theory Index for urban infrastructure investment in Sub-Saharan Africa. Int J Manag Data Anal 6, 354–388. · Marolla C. (2026b). The Risk Mechanism Theory Index as a triage and evidence-grading layer for hepatocellular carcinoma surveillance. SSRN (preprint).